



SIMATS
ENGINEERING



SIMATS
Sreeetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

AI-Based Student Performance Prediction System

CSA 1711- Artificial Intelligence

GUIDED BY:

Dr. Krishnakumar Kathiresan

PRESENTED BY:

P. Shashank

T. Niranjana Kumar

P. Abhinay



Abstract



- AI-based system for predicting student academic performance.
- Uses academic and behavioral student data.
- Applies Random Forest and XGBoost algorithms.
- Compares models using actual evaluation metrics.
- Identifies weak/at-risk students at an early stage.
- Provides performance prediction, risk level and contributing factors.



Introduction



- Student performance is influenced by multiple academic and behavioral factors.
- Manual identification of weak students can be difficult and time-consuming.
- Machine Learning can identify patterns from historical student data.
- The system predicts performance for new students.
- Web dashboard provides easy visualization and analysis.
- Supports individual and batch student prediction.

Problem Statement

- Student performance is difficult to monitor manually.
- Weak students may not be identified early.
- It is difficult to accurately predict a student's performance from available academic data.
- Factors like attendance, marks, assignments, and study hours affect student performance.
- An AI-based system is needed to predict performance and identify at-risk students early.

Literature Review

Year	Approach	Method Used	Advantage	Limitation	Year
2022	Student Performance Analysis	Data Mining	Identifies academic patterns	Limited prediction	2022
2023	Academic Prediction	Classification ML	Automated prediction	Depends on data quality	2023
2023	Student Risk Detection	Decision Trees	Easy interpretation	Limited generalization	2023
2024	Performance Prediction	Random Forest	Good classification	Requires tuning	2024

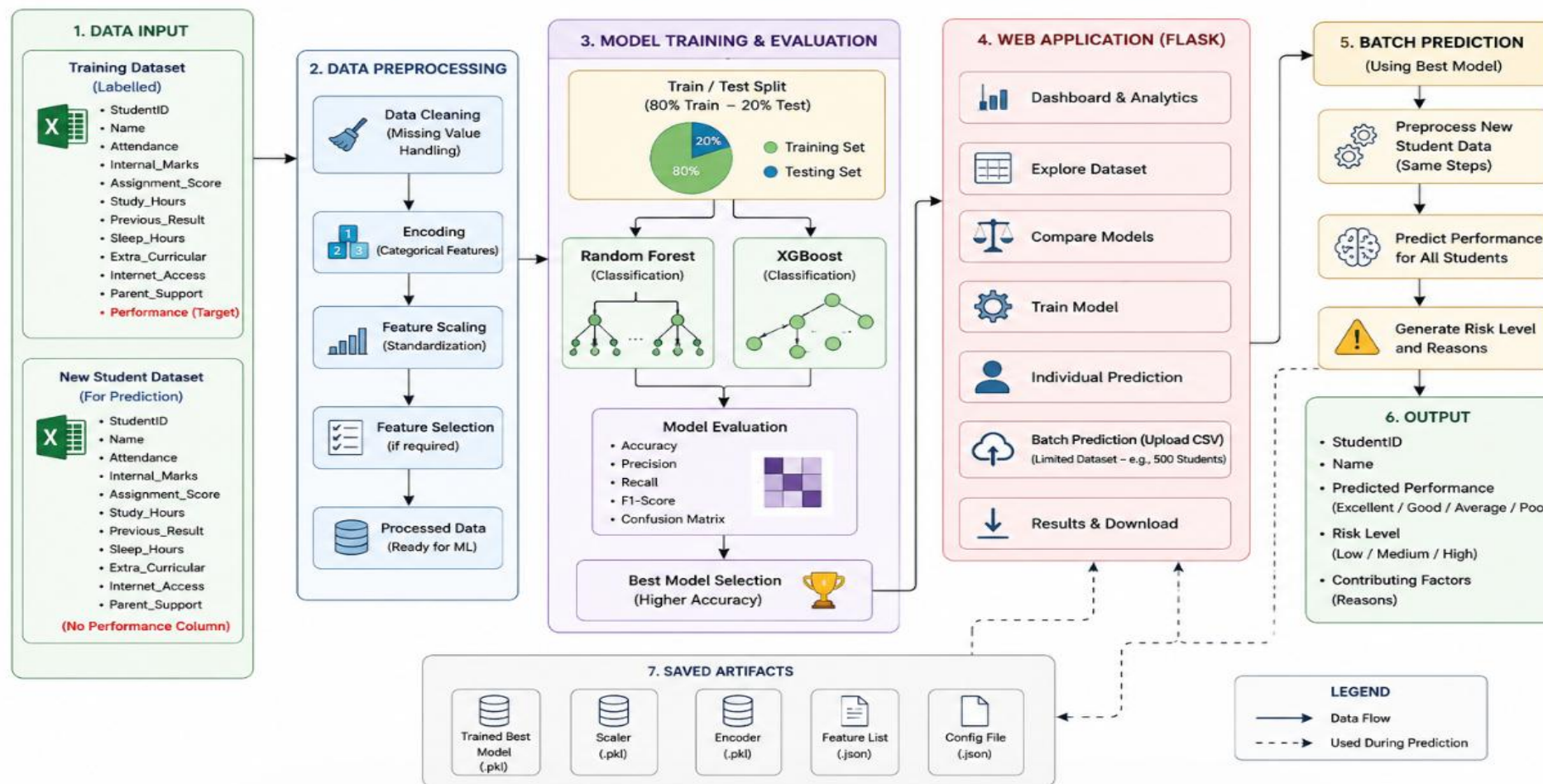


Engineer to Excel



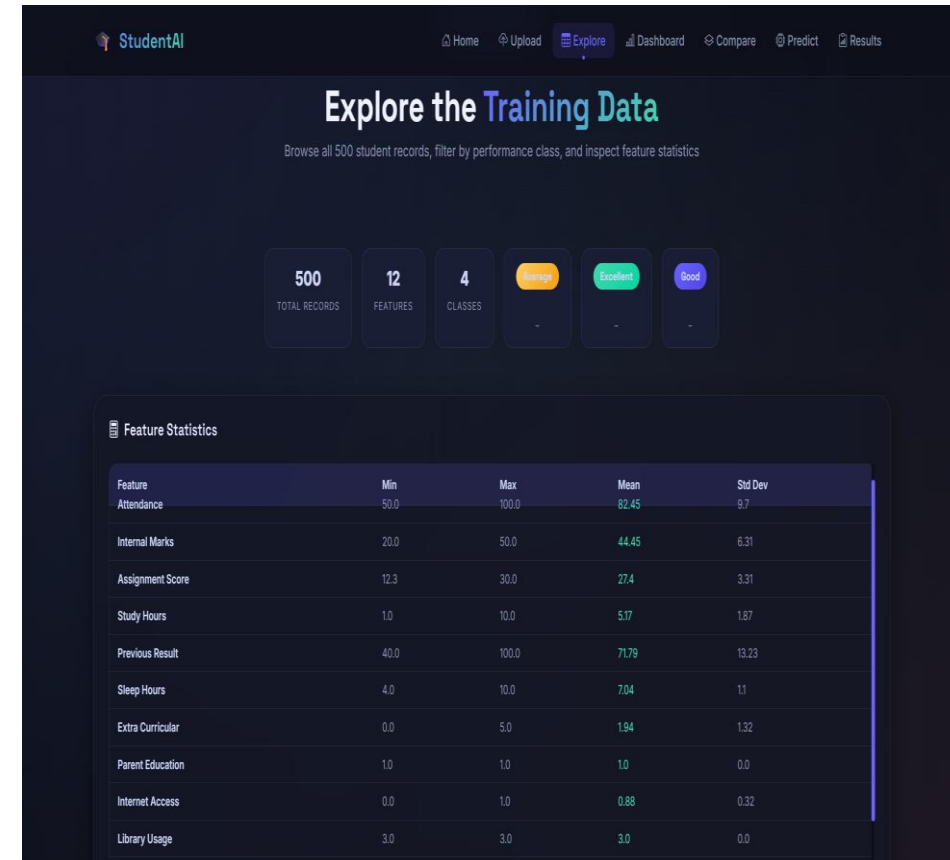
SYSTEM ARCHITECTURE

AI-Based Student Performance Prediction System



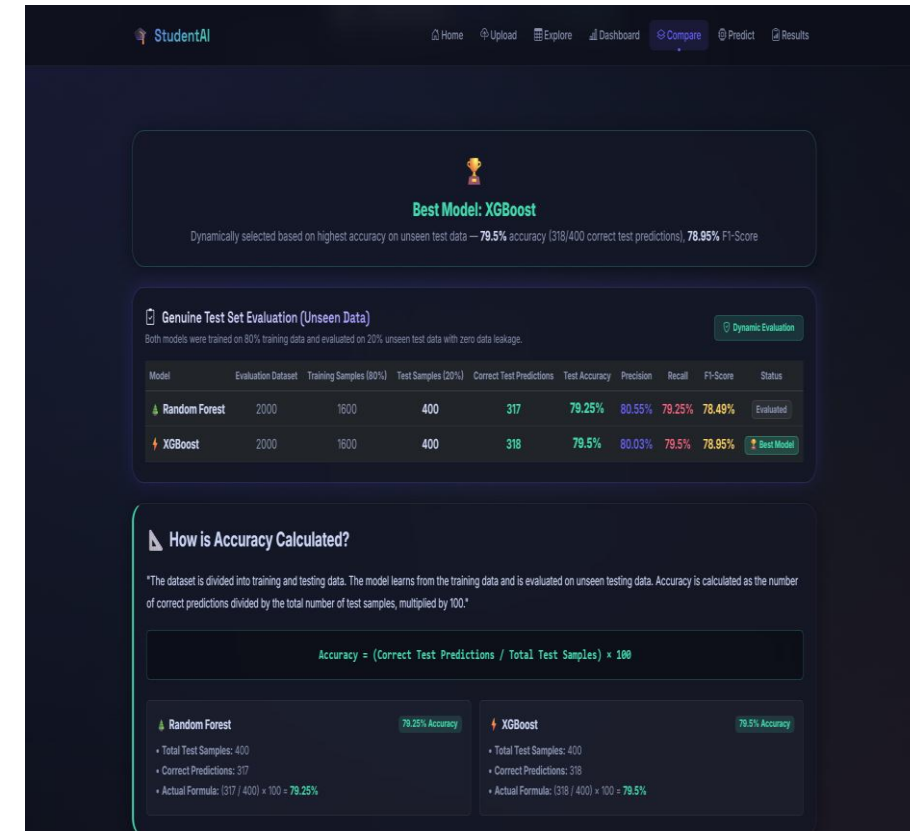
Module 1: Data Collection & Preprocessing

- Collect student academic and behavioral data.
- Validate dataset columns.
- Handle missing values.
- Encode categorical data.
- Separate Student ID and Name from ML features.
- Prepare data for machine learning.



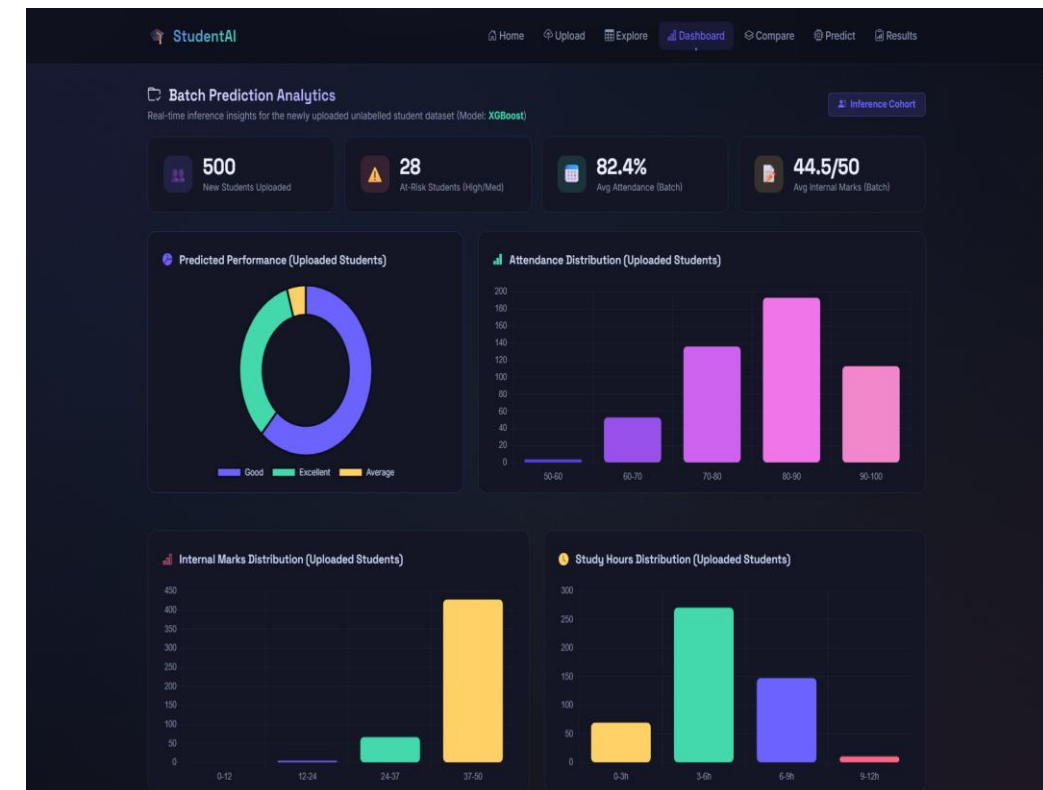
Module 2: Model Development

- Split labelled dataset into training and testing data.
- Train Random Forest model.
- Train XGBoost model.
- Generate predictions on unseen test data.
- Calculate Accuracy, Precision, Recall and F1 Score.
- Compare models.
- Select the best-performing model.
- Save the trained model.

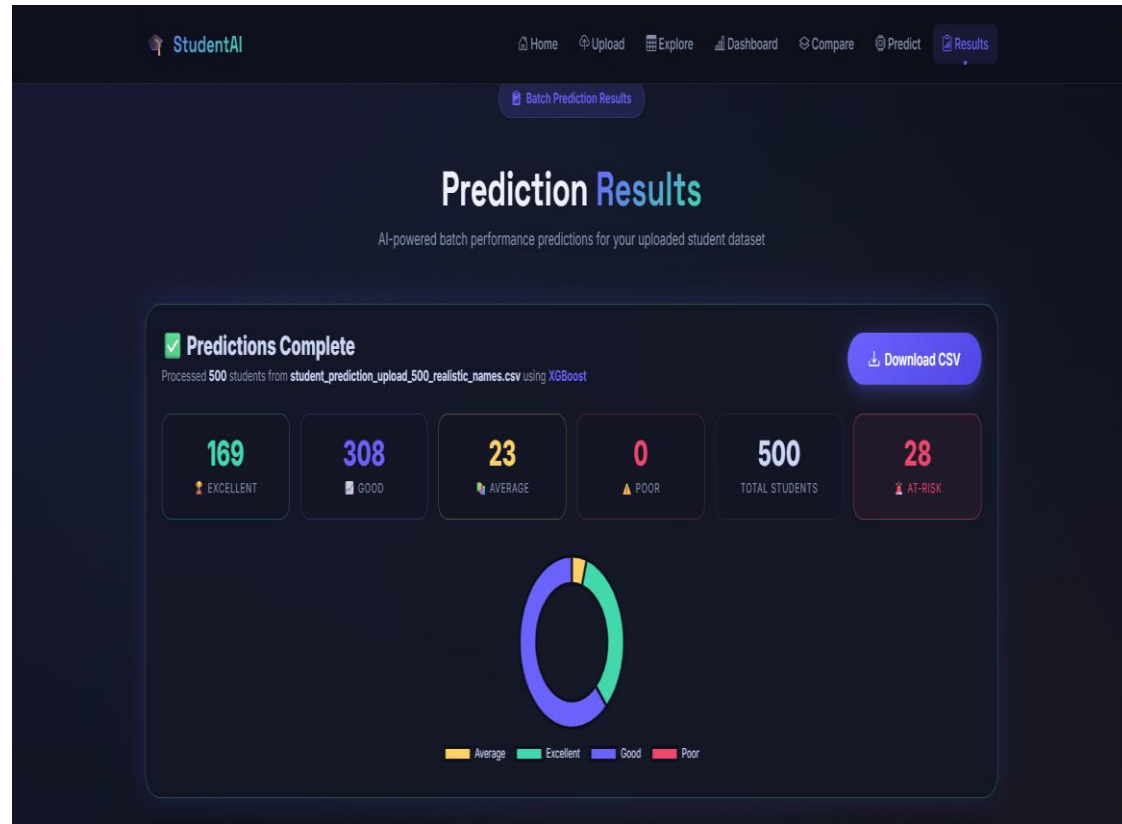


Module 3: Dashboard & Deployment

- Provides web-based interface.
- Dataset exploration.
- Model comparison.
- Analytics dashboard.
- Individual student prediction.
- Batch prediction for students.
- Results and downloadable prediction



Output



Prediction Results (showing first 100 of 500)

All Categories

Export All

#	StudentID	Name	Attendance	Internal Marks	Assignment Score	Study Hours	Previous Result	Sleep Hours	Extra C	Education	Int
1	S001	Arjun Reddy	77.7	50.0	23.9	3.7	70.7	4.8	3.0		1.0
2	S002	Rahul Naidu	68.1	50.0	30.0	7.7	62.4	7.4	2.0		1.0
3	S003	Priya Mishra	85.1	50.0	28.1	3.9	87.8	8.0	1.0	1.0	1.0
4	S004	Ananya Agarwal	81.9	50.0	30.0	7.6	91.2	7.5	0.0	1.0	1.0
5	S005	Kiran Bhat	96.5	50.0	30.0	4.5	69.5	7.5	0.0	1.0	1.0

Legend: Excellent (orange), Good (blue), Average (green), Poor (yellow), High Risk (red), Medium Risk (orange)

Upload New Dataset View Dashboard Explore Dataset

Applications

- **Early Identification of Weak Students** – Identifies low-performing students at an early stage.
- **Academic Performance Prediction** – Predicts student performance using academic and behavioral factors.
- **Teacher Decision Support** – Helps teachers analyze and monitor student performance.
- **Student Risk Assessment** – Classifies students into low, medium, and high-risk categories.
- **Academic Intervention Planning** – Helps provide timely support and guidance to at-risk students.
- **Batch Student Analysis** – Predicts the performance of multiple students simultaneously.

Future Scope

- **Real-Time Data:** Integrate live attendance and academic data into the system.
- **Personalized Support:** Provide suggestions based on each student's performance.
- **Explainable AI:** Show the main factors behind each prediction.
- **Performance Tracking:** Track student performance across different semesters.
- **Mobile Application:** Make the system accessible through a mobile app.
- **Cloud Deployment:** Deploy the system online for easy access and scalability.

Conclusion

- ✓ The system predicts student academic performance using Machine Learning.
- ✓ Random Forest and XGBoost are used and compared for prediction.
- ✓ The system identifies weak and at-risk students at an early stage.
- ✓ It provides the important factors affecting student performance.
- ✓ The web dashboard makes student analysis and prediction simple and effective.

References

- Breiman, L. (2001). Random Forests. *Machine Learning*, 45, 5–32. Source: Springer.
- Chen, T. & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. Source: ACM.
- Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830. Source: JMLR.
- McKinney, W. (2010). Data Structures for Statistical Computing in Python. *Proceedings of the 9th Python in Science Conference*. Source: SciPy.
- Abadi, M. et al. (2016). TensorFlow: A System for Large-Scale Machine Learning. *OSDI*. Source: USENIX.

Thank You!

For your attention

